25900416 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEGEO200 - ENDOGENOUS EARTH PROCESSES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer in 60 words

Answer any **5** questions

Each question carries **2** marks

1. State the age of the Earth. [U] / [CO1]
2. List the three major types of plate boundaries associated with tectonic activity. [K] / [CO3]
3. Evaluate the size range of different pyroclastic materials. [E] / [CO4]
4. Mention which physiographic region of India is at the greatest risk of high-magnitude earthquakes. [E] / [CO5]
5. Define transform boundaries of tectonic plates. [K] / [CO3]
6. Name one catastrophic event that could alter Earth’s geology. [U] / [CO2]

[2x5 = 10]

Part B

Short notes in 250 words

Answer any **6** questions

Each question carries **8** marks

7. Discuss how the concepts of isostasy help to explain the post-glacial rebound of different regions on Earth. [A] / [CO3]

8. Evaluate the difference between pahoehoe and Aa lava flows. [E] / [CO4]

9. Explain the role of faults in the mechanism of earthquakes as described by the Elastic Rebound Theory. [U] / [CO5]

10. DFiscxuss how the Japan type of orogeny different from the Andes type in terms of plate interactions [U] / [CO3]

11. Discuss geological similarities across continents as proof of continental drift. [U] / [CO2]

12. Explain how the seafloor spreading fuels the plate tectonic theory? [A] / [CO3]

13. Discuss shield volcano, its size, shape and composition with neat diagram. [E] / [CO4]

14. Discuss the differences between tectonic and volcanic earthquakes. Describe their causes, typical locations, and key characteristics. [U] / [CO5]

[8x6 = 48]

Part C

Essay in 400 words

Answer any **1** questions

Each question carries **12** marks

15. Explain the concept of plate tectonics, and discuss the landforms associated with various plate margins [U] / [CO3]

16. Critically assess the overall contributions and limitations of different earthquake measurement techniques in studying earthquakes. [E] / [CO5]

[12x1 = 12]